

Gwen MAUDET, Post Doctoral Researcher



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Introduction

I am currently a postdoctoral researcher at the University of Luxembourg, where my work focuses on integrating machine learning (ML) techniques to enhance the efficiency of solvers for Mixed Integer Linear Programming (MILP) problems. I previously completed my PhD at IMT Atlantique in Rennes, France, where my research centered on data transmission strategies for large-scale, highly constrained sensor networks.

For a quick and structured overview of my profile, please visit my [professional website](#).

Professional Experience

- December 2023 – Present
 - At *University of Luxembourg, Esch-sur-Alzette, Luxembourg*. **Postdoctoral Researcher in Machine Learning for Mixed Integer Linear Programming**. Funded by the collaborative project Ultra Big Optimization.
Worked on the integration of learning techniques inside key components of exact solvers. In particular, I investigate population-based algorithmic strategies and the generalization of these methods across a broad spectrum of mixed integer linear programming instances. In parallel, I have delivered several lectures at the master's level and supervised multiple students on course projects and in their roles as student assistants. I currently co-supervise a PhD student whose research is also part of the Ultra Big Optimization project.
- April - September 2020
 - At *INRIA, Rennes, France*. **Intern in Detecting Bias in Search Engines**.
Contributed to the development of tools for detecting biases between search engines, including their integration into the web platform SNIDE and their evaluation through extensive experimental studies presented in a journal publication.
- June - August 2019
 - At *Acklio, Rennes, France*.
Intern in Header Compression for IoT Networks Using Clustering.
Developed a clustering-based method to identify similarities among IoT transmission headers, enabling the automatic generation of compression rules based on the mechanism defined in RFC 8724.

Education

- 2020 - 2023 • At *IMT Atlantique, Rennes, France*. PhD in **Exploiting Sensor Similarity to Enhance Data Collection in Massive IoT Networks**. Funded by the [VaLaDOE research chair](#).
Focused on the monitoring of physical phenomena in environments equipped with large numbers of highly constrained sensors, both in terms of energy and communication capabilities. In particular, I investigated load-balancing mechanisms and similarity-based sensor grouping strategies under highly dynamic conditions.
Defense Date: November 23, 2023; [video](#).
Jury Composition: Prof. Laurent Toutain, IMT Atlantique (*Director*) - Prof. Mireille Batton-Hubert, École des Mines de Saint-Étienne (*Co-director*) - Prof. Patrick Maillé, IMT Atlantique (*Advisor*) - Prof. Alexandre Guitton, Université Clermont Auvergne (*Reviewer*) - Dr. Julien Montavont, Université de Strasbourg (*Reviewer*) - Dr. Kinda Khawam, Université de Versailles.
- 2016 - 2020 • At *IMT Atlantique, Nantes, France*. General engineering student in **computer science for decision support**.
Completed courses in machine learning, operational research, and graph theory. Participated in a one-semester academic exchange at Universiti Teknologi Petronas (Malaysia).
- 2014 - 2016 • At *Rabelais, Saint-Brieuc, France*. Student in **Classes préparatoires aux grandes écoles**.
Completed a two-year intensive program in mathematics and physics ([MPSI-MP](#)) to prepare for the national competitive exam for entry into highly ranked engineering schools.

Research Writings

PhD Thesis

- **G. Maudet**, Exploiting Sensor Similarity to Enhance Data Collection in Massive IoT Networks, [Link](#).

International Journals

- P. Maillé, **G. Maudet**, M. Simon, and B. Tuffin, "Are Search Engines Biased? Detecting and Reducing Bias using Meta Search Engines," en, *Electronic Commerce Research and Applications*, p. 101 132, Feb. 2022, issn: 1567-4223. (SJR: Q1) [Link](#).

International Conferences

- A. Vorokhta, **G. Maudet**, G. Danoy, "Selecting the Best Lower-Bound Strategy in a Branch-and-Bound Algorithm Using Genetic Programming", *International Workshop on Big Optimization*, 2026, [Link](#).
- **G. Maudet**, G. Danoy, "A Distance Metric for Mixed Integer Programming Instances," in *European Conference on Artificial Intelligence*, 2025, (CORE: A) [Link](#).
- **G. Maudet**, G. Danoy, "Search Strategy Generation for Branch and Bound Using Genetic Programming," in *Proceedings of the AAAI Conference on Artificial Intelligence 39*, 2025, (CORE: A*) [Link](#).
- **G. Maudet**, M. Batton-Hubert, P. Maille, and L. Toutain, "Energy Efficient Message Scheduling with Redundancy Control for Massive IoT Monitoring," in *IEEE Wireless Communications and Networking*, 2023, (CORE: B) [Link](#).

Research Writings (continued)

- **G. Maudet**, M. Batton-Hubert, P. Maille, and L. Toutain, "Emission Scheduling Strategies for Massive-IoT: Implementation and Performance Optimization," in *IEEE/IFIP Network Operations and Management Symposium*, Apr. 2022, (CORE: B) [Link](#).

National Conferences and Extended Abstracts

- M. Da Cunha, **G. Maudet**, G. Danoy, "Cutting Plane Selection for Mixed-Integer Linear Programming Using Genetic Programming," Extended Abstract, in *Conference on Optimisation and Learning*, April 2026.
- B. Blé, **G. Maudet**, G. Danoy, "Distance-based Portfolio Learning for Mixed-Integer Programming," Extended Abstract, in *Conference on Optimisation and Learning*, April 2026.
- **G. Maudet**, G. Danoy, "Une Distance pour les Instances de Programmation Linéaire en Nombres Entiers Mixtes," in *Recherche opérationnelle et aide à la décision en France*, February 2026.
- **G. Maudet**, G. Danoy, "A Distance Metric for Mixed Integer Programming Instances," in *Operations Research BELgium*, February 2026.
- A. Vorokhta, **G. Maudet**, G. Danoy, "Generating LB Computation Strategies in a B&B Algorithm: Application to the PFSP," Extended Abstract, in *Conference on Optimisation and Learning*, April 2025.
- **G. Maudet**, M. Batton-Hubert, P. Maille, and L. Toutain, "Grouper les Capteurs Similaires Grace à leurs Données dans le Contexte de Massive IoT," in *26èmes Rencontres Francophones sur les Aspects Algorithmiques des Télécommunications*, May 2024.
- **G. Maudet**, M. Batton-Hubert, P. Maille, and L. Toutain, "Réduction de la Redondance de Messages des Capteurs dans un Contexte Massive IoT," in *25èmes Rencontres Francophones sur les Aspects Algorithmiques des Télécommunications*, May 2023.
- **G. Maudet**, M. Batton-Hubert, P. Maille, and L. Toutain, "Développement de nouvelles stratégies de monitoring grâce a une approche Massive IoT", *Platform IA*, June 2022.

Preprints and Ongoing Work

- "Adaptive Lower Bound Evaluation for the Permutation Flowshop Scheduling Problem," in collaboration with Alisa Vorokhta. *Under review*.
- "SkyHOSTop: Cost-Aware Multi-Cloud Streaming Overlay Optimization," in collaboration with Muhammad Arslan Tariq. *Under review*.
- "Cutting Plane Selection for Mixed-Integer Linear Programming Using Genetic Programming," in collaboration with Mathis Da Cunha. *Under review*.
- "Performance Driven Portfolio Approach to Generalize Machine Learning for Mixed Integer Programming." *In progress*.
- "Distance Based Clustering of the Mixed-Integer Programming Space for Portfolio-Based Machine Learning Generalization," in collaboration with Baka Junior Cedric Blé. *In progress*.

Research Writings (continued)

- **G. Maudet**, M. Batton-Hubert, P. Maille, and L. Toutain, “Grouping Sensors Based on Observations in a Massive IoT Deployment,” *Preprint*. [Link](#).

Supervision

PhDs

- 2026–Present
- **Simon Renard**. Research on enhancing solver performance through machine learning.
 - **Muhammad Arslan TARIQ**. Research on streaming through multi cloud multi path connections: infrastructure and path optimisation.
- 2023–Present
- **Alisa Vorokhta**. Research on the application of machine learning methods reducing the impact of the lower bound computation when solving MILP using branch and bound.

Interns

- 2025
- **Baka Junior Cedric Ble**, *Master’s student* (5 months). Worked on clustering methods for MILP instances, aiming to develop a portfolio-based approach where specific MLMILP models are applied to homogeneous groups of instances.
 - **Mathis Da Cunha**, *Master’s student* (5 months). Research on the use of genetic programming to define cutting plan selection strategies within branch-and-bound frameworks.
- 2022
- **Issam Belhorma**, *Master’s student* (4 months). Applied kriging-based techniques to estimate sensor positions using returned sensor data and references from known-position nodes.
- 2021
- **Carlos Delgado**, *Master’s student* (3 months). Worked on sensor deployment in vertical green walls for monitoring and regulating soil humidity levels.

Student jobs

- 2025
- **Diptaraj Sen**, *Master’s student* (5 months, 5 hours per week). Evaluating an heterogeneous benchmark of instances considering an alternative distance metric between instances.
- 2024
- **Daniele Ferrario**, *Master’s student* (3 months, 5 hours per week). Understanding a scientific paper and the code that defines a reinforcement learning method for cutting plan selection.

Teaching

- 2024-2025
- Project Lead for student groups at *University of Luxembourg, Esch-sur-Alzette, Luxembourg*. Supervised teams of three master's students in Information and Computer Science, working on research projects for four hours per week over the summer semester.
- 2025
- Evaluating the influence of the cutting plan strategy on the solving performance.
- 2024
- Evaluating the performance of two lower-bound strategies in the branch-and-bound applied to the Permutation Flowshop Scheduling Problem.
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- Lecturer at *University of Luxembourg, Esch-sur-Alzette, Luxembourg*. Taught the winter semester of the first-year master's program in Information and Computer Sciences in an amphitheater. Delivered 4 sessions for 58 students in 2024 and 6 sessions for 62 students in 2025, each lasting 1.5 hours:
 - Introduction to Integer Linear Programming: integer linear programming, branch-and-bound ([Slides](#)).
 - Elements of the branch and bound: definitions of different branching and searching strategies ([Slides](#)).
 - Population-Based Metaheuristic Algorithms: evolutionary algorithms and swarm intelligence methods ([Slides](#)).
 - The Traveling Salesman Problem in a Jupiter notebook: brute force, Integer linear programming resolution, heuristics, and metaheuristics comparison ([exercise – with correction](#)).
 - Ongoing Research: hybridization of the branch-and-bound and genetic programming.
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- 2022 - 2023
- Practice teacher at *IMT Atlantique, Rennes, France*. Supported the first semester of a two-year international master's program in IT, focusing on mathematics and programming foundations. Worked with 12 students in 2022 and 24 in 2023.
 - Matlab - 6h (2022) + 6h (2023): applications to complex numbers, matrix calculations, sequences, and signal processing.
 - Algebra - 6h (2022) + 6h (2023): from linear space definition to eigenvectors.
 - Introduction to Python - 9h (2022) + 9h (2023): lists, functions, focus on *RegEx*, *Numpy*, and *Pandas* libraries.
 - Python for Data Science - 16h (2022) + 16h (2023): from problem definition to supervised and unsupervised algorithm principles. Project: real world data science problem (2022), presentation of a new ML concept (2023).
 - Operational Research - 7h30 (2022): linear programming, integer linear programming, and heuristics. Project: real world problem solved using exact and approximate methods.
 - Probability and Statistics - 4h (2022): generating a random variable from cumulative distribution functions.
 - Signal Processing - 2h30 (2023): harmonics detection, trilateration object positioning.

Teaching (continued)

- 2012 - 2021
- Private tutor in *Brittany, France*. Provided personalized teaching to primary and secondary school students, averaging two students per week.

Review Committee

PC chair

- 2025-2026
- Shadow PC Chair for the *Algotel-Cores* Conference. Led the organization of the shadow reviewing process, including reviewer recruitment, training sessions, bidding setup, and paper discussions, enabling early-career researchers to engage in a full peer-review cycle without affecting official decisions.

Reviewing

- Two times reviewed two seven pages double column papers for the *PDCO* Workshop of the international conference *IPDPS*.
- 2024
- Reviewed two papers for the international journal *IEEE TNSM*.
 - Reviewed one fourteen pages double column paper for the international journal *IEEE Sensors Journal*.
- 2023
- Shadow reviewed four four-page single-column conference papers for the French national conference "*Cores et Algotel*".

Scientific Talks

- 2026
- **Une Distance pour les Instances de Programmation Linéaire en Nombres Entiers Mixtes**, at *ROADEF, Tour, France*.
 - **A Distance Metric for Mixed Integer Programming**, at *ORBEL, Leuven, Belgium*.
- 2025
- **A Distance between Mixed Integer Programming Instances: on the Road to Generalizability of Machine Learning Solving Strategies**, at *ISyE monthly Seminars, UGent, Gent, Belgium* (Seminar webpage).
 - **A Distance Metric for Mixed Integer Programming**, at *ECAL, Bologna, Italy*.
 - **Portfolio Approaches for the Generalizability of ML Solving Strategies in MIP Solvers** at *ZIB seminars, Zuse Institute Berlin, Berlin, Germany* (seminar webpage).
 - **A Distance between Mixed Integer Programming instances: on the Road to Generalizability of Machine Learning Solving Strategies**, at *DSD weekly Seminars, IMT Atlantique, Brest, France*. (departement webpage).
 - **Search Strategy Generation for Branch and Bound Using Genetic Programming**, at *AAAI, Philadelphia, USA*. Oral presentation and poster. The oral presentation was one of approximately 600 selected from over 3,000 accepted papers at the conference.
- 2024
- **Grouper les capteurs similaires grâce à leurs données dans le contexte de Massive IoT**, at *ALGOTEL, Saint-Briac-sur-Mer, France*.
- 2023
- **Energy Efficient Message Scheduling with Redundancy Control for Massive IoT Monitoring**, at *IEEE WCNC, Glasgow, Scotland*.

Scientific Talks (continued)

- **Réduction de la Redondance de Messages des Capteurs dans un Contexte Massive IoT**, at *ALGOTEL, Cargese, France*.
- 2022 • **New Monitoring Strategies using Massive IoT**, at *Plate-forme Intelligence Artificielle, Saint-Étienne, France*.
- **Emission Scheduling Strategies for Massive-IoT: Implementation and Performance Optimization**, at *IEEE NOMS, Budapest, Hungary*.
- 2021 • **Poster: Dynamic management of a field of sensors to extend the monitoring time**, at *Symposium IMT: internet du futur, Villeneuve d'Ascq, France*.
- **Strategies for transmitting LoRa wireless sensors to optimize supervision**, at *Journées LP-WAN, Clermont-Ferrand, France*.
- 2019 • Master of ceremonies, at *IMT Atlantique students' graduation ceremony, Nantes, France*.

Miscellaneous

Position in Associations and Institutions

- 2022 - 2023 • **President of the association of PhD students and young researchers** in *IMT Atlantique Rennes*.
- 2023 • **Representative of the PhD students** for the *doctoral school "SPIN"*.

Volunteering

- 2022 • **Conference Volunteer** at the national conference *Plateforme Intelligence Artificielle*.

Languages

- French Native, English C1, Spanish B1, Italian B1

Sports

- Indoor and outdoor climber, rope and boulder. Trail and road runner, mountain hiker, and cyclotourist biker. Black belt Judoka.

References

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